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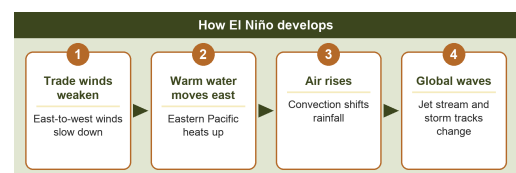

Unraveling El Niño: Ocean-Atmosphere Interactions and Global Climate Impacts

Abstract

In late 2015, the world was experiencing only the third super El Niño on record. Climate scientist Kevin Trenberth explained what El Niño is, why it happens, and how it reshapes weather around the globe, from typhoons in the Pacific to floods in India and rain in California. He also explained how global warming amplifies El Niño's effects.

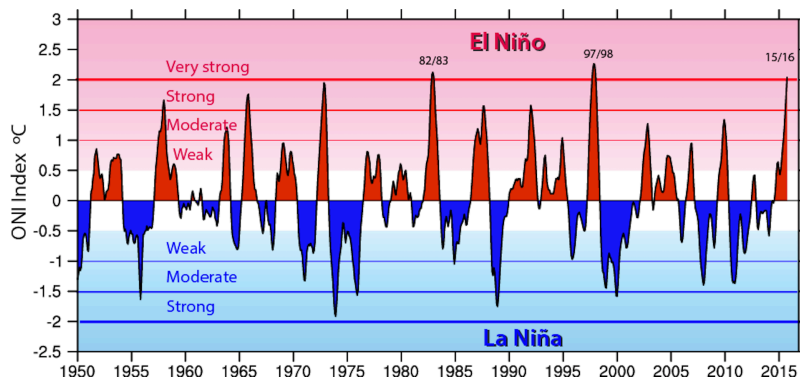
Introduction

Every three to seven years or so, surface waters across the tropical Pacific Ocean, from the International Dateline to the coast of South America, become extremely warm. This is El Niño. Its name, Spanish for the boy Christ child, originally referred to warm water that appeared off Peru near Christmas. Today it describes changes across the entire Pacific basin. El Niño alternates with La Niña, its cool opposite, and is linked to a see-saw in air pressure called the Southern Oscillation. Scientists call the whole system the El Niño-Southern Oscillation, or ENSO. Before 2015, there had been only two super El Niños: in 1982-83 and 1997-98.



How Do Scientists Study This?

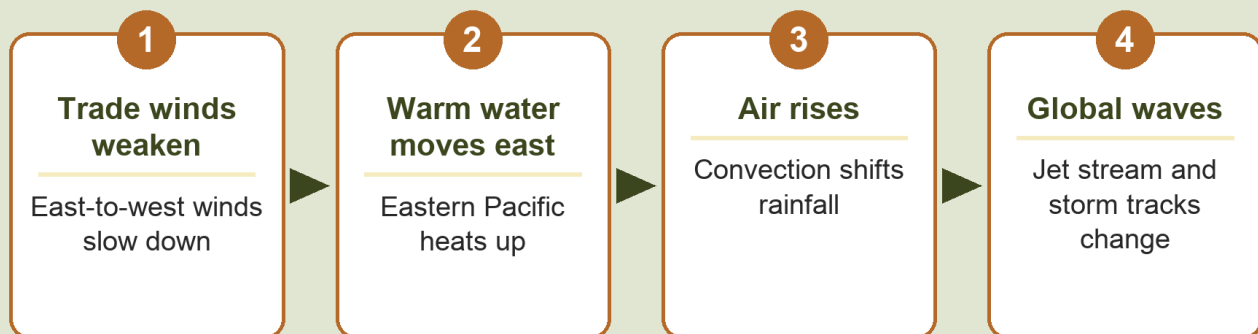
Scientists track El Niño with measurements of sea surface temperature across the tropical Pacific, summarized in the Oceanic Niño Index (ONI), which classifies warm and cold phases as weak, moderate, strong, or very strong. They study how the ocean and atmosphere interact: changing sea temperatures alter winds and rainfall, which then change ocean currents and temperatures, creating a feedback loop. Evidence from coral reefs and Andean glacier ice shows ENSO has occurred for thousands of years. To predict impacts, scientists study teleconnections, the waves in the atmosphere that carry El Niño's influence to distant regions, and compare the current event with past El Niños.



FROM THE ARTICLE

The Oceanic Niño Index shows warm (red) El Niño and cold (blue) La Niña phases since 1950. Image: NCAR, author provided.

How El Niño develops



Study the Process

Measure the ocean

Sea surface temperatures across the tropical Pacific are tracked with the Oceanic Niño Index.

Trace the feedback

Scientists follow how warm water, weaker trade winds, and rainfall reinforce one another.

Predict impacts

Teleconnections and past events help forecast where floods, droughts, and storms will occur.

What Did They Discover?

During El Niño, the trade winds weaken, and warm water surges east along the equator, raising sea level in the east by as much as a foot. Warm, moist air rises, shifting rainfall and creating atmospheric waves that bend the jet stream. The 2015 El Niño brought a record 25 category 4 and 5 hurricanes and typhoons in the northern Pacific, beating the previous record of 18, and drought fueled wildfires in Indonesia. A very warm Bay of Bengal brought 47 inches of rain to Chennai, India, in November. For the U.S., scientists expected heavy rain across the South, possible flooding in California, and mild, dry weather in the North. Global warming sets the background, making 2015 the warmest year on record and amplifying El Niño's impacts.

The 2015 super El Niño

3rd

super El Niño on record, after 1982-83 and 1997-98

25

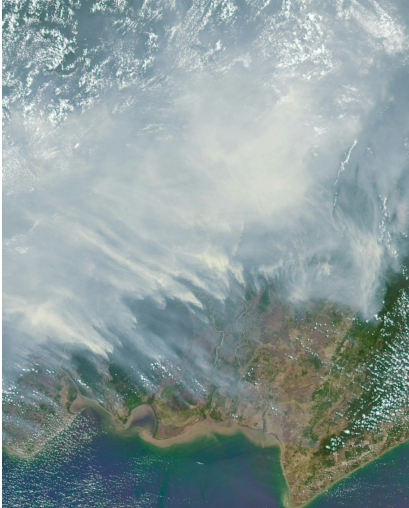
category 4-5 Pacific storms, beating the record of 18

47 in

of November rain in Chennai, India

When global warming and El Niño push in the same direction, they have the biggest effects and records are broken.

Look at the Evidence



FROM THE ARTICLE

Drought fuels wildfires on Borneo, October 14, 2015, a predictable impact of El Niño. Image: NASA/GSFC/LaRC/JPL-Caltech, MISR Team.

Takeaways

Ocean and air together

El Niño arises from feedback between Pacific Ocean temperatures, winds, and rainfall.

Global reach

Atmospheric waves spread El Niño's effects to weather far from the Pacific.

Warming amplifies

Climate change adds heat and moisture, intensifying El Niño's floods, droughts, and storms.

Key Terms

El Niño

A natural warming of surface waters across the central and eastern tropical Pacific Ocean that affects weather worldwide.

La Niña

The cool counterpart of El Niño, when tropical Pacific surface waters become cooler than normal.

ENSO

The El Niño-Southern Oscillation, the combined ocean and atmosphere system that includes El Niño, La Niña, and pressure changes.

Convection

The rising of warm, moist air, which forms clouds and rain and releases heat into the atmosphere.

Oceanic Niño Index (ONI)

A measure of tropical Pacific sea surface temperature anomalies used to identify and rank El Niño and La Niña events.

Match-Up: draw a line from each term to its clue

A El Niño

B La Niña

C ENSO

D Convection

E Oceanic Niño Index (ONI)

1 A measure of tropical Pacific sea surface temperature anomalies used to identify...

2 The El Niño-Southern Oscillation, the combined ocean and atmosphere system that includes...

3 The rising of warm, moist air, which forms clouds and rain and...

4 The cool counterpart of El Niño, when tropical Pacific surface waters become...

5 A natural warming of surface waters across the central and eastern tropical...

Answer key: A-5 · B-4 · C-2 · D-3 · E-1

Activity

El Niño World Map

Students map El Niño's global impacts and connect them to ocean-atmosphere changes in the Pacific.

Materials: Blank world map; Colored markers: red for dry, blue for wet, orange for storms; Lesson text

1. Draw the tropical Pacific and shade the area of unusually warm water during El Niño.
2. Mark regions that become drier: Australia, Indonesia, Southeast Asia, the Philippines, Hawaii, and northeastern Brazil.
3. Mark regions that become wetter, including the southern United States and the west coast of South America.
4. Add the 2015 events from the lesson: Pacific typhoons, Indonesian wildfires, and Chennai floods.
5. Write a paragraph explaining how warm Pacific water can cause floods and droughts thousands of miles away.

References

Trenberth, K. (2015). What North America can expect from El Niño. The Conversation.
<https://doi.org/10.64628/AAI.4f4ve77g6>